



Assessing Vertical Ground Motion Along the New Jersey Shoreline with Continuous GPS

Sirichenko, D.¹ & Marsellos, A.E.²

¹The Pingry School, 131 Martinsville Road, Basking Ridge, NJ, USA

²Dept. Geology, Environment and Sustainability, Hofstra University, Hempstead, NY, USA



Abstract:

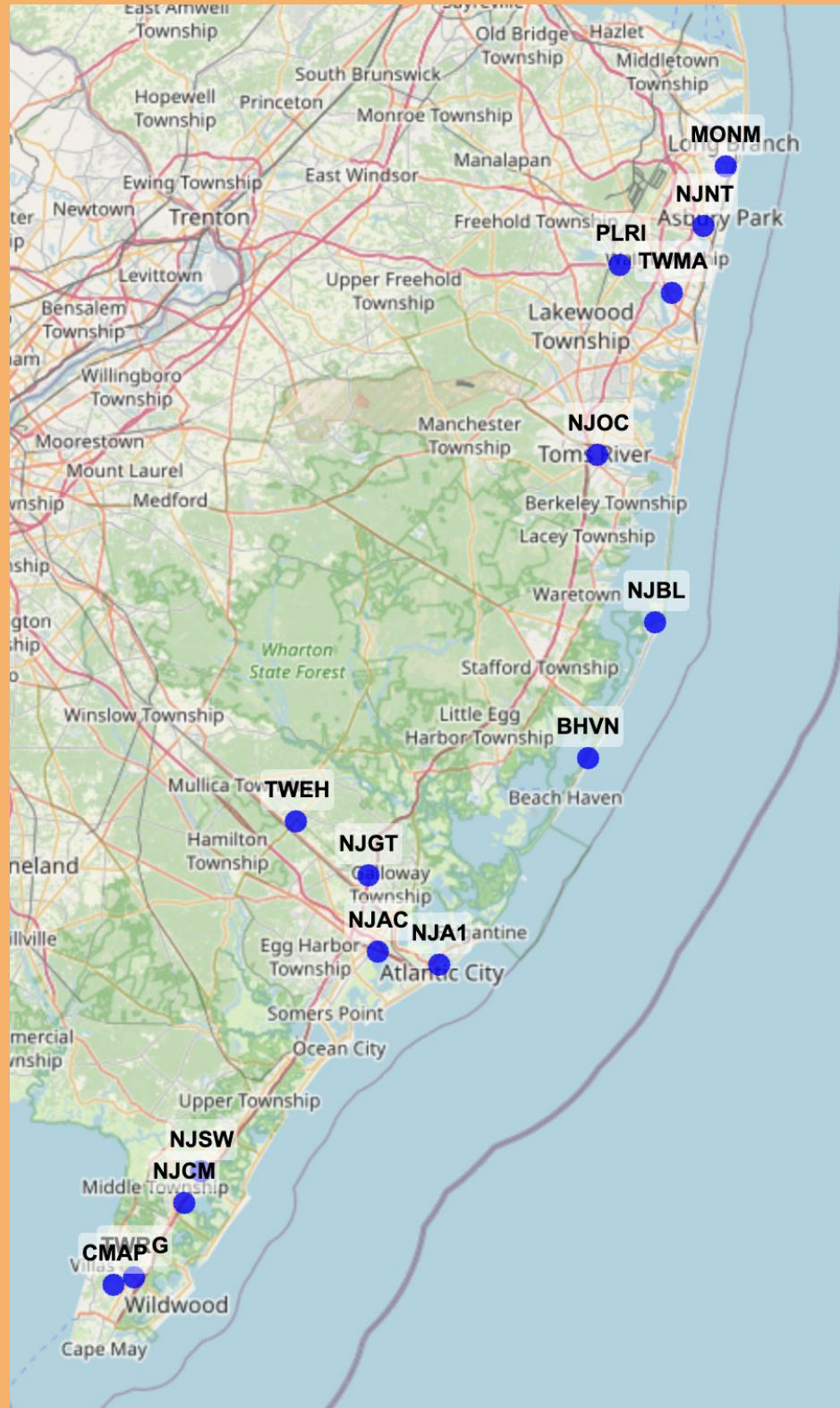
Coastal regions of New Jersey face compounded risks from both sea-level rise and land subsidence, yet the spatial variability of vertical ground motion along the shoreline remains poorly constrained. In this project, continuous GPS (cGPS) stations were analyzed to quantify horizontal and vertical displacement trends and assess their implications for coastal stability. Using R-based processing scripts, we downloaded, cleaned, and visualized multi-year displacement records (east, north, and up components) from shoreline GPS stations.

Outlier removal and trend analysis revealed consistent regional northward and westward motion, with notable variability in the vertical (Up) component. On average, the GPS stations exhibit subsidence of approximately 2.45 mm per year, consistent with glacial isostatic adjustment and sediment compaction typical of the Mid-Atlantic coastal plain. Several stations showed more pronounced subsidence, suggesting an increased vulnerability to coastal flooding, while one station displayed apparent uplift likely related to data discontinuity.

By integrating displacement rate mapping with geospatial analysis, this study improves understanding of localized deformation and contributes to long-term assessments of shoreline resilience in the context of sea-level rise and climate change.

Background Information:

- Sea level rise threatens coastal New Jersey communities
- Understanding vertical land motion (subsidence/uplift) is critical for accurate sea level rise projections and coastal planning
- GPS stations provide continuous, precise measurements of land elevation changes
- GPS vertical displacement data contains multiple signal sources: long-term trends (subsidence), seasonal atmospheric loading (annual/semi-annual cycles), and noise
- Separating these signals requires careful data processing and quality control



GPS stations

The GPS data used in this analysis was from multiple stations along the New Jersey shore. Blue dots represent the stations.

Figure 1: Location of GPS stations used along the New Jersey shoreline.

Method/Materials:

- Nevada Geodetic Laboratory**
 - GPS data from GPS stations in New Jersey
 - Up, North, East data
- RStudio (Version 2024.04.2+764)**
 - Extrapolated data from NGL
 - Fill in missing data (gaps in NGL graphs)
 - Packages used: data.table, imputeTS, etc.
 - Moving average (kza)
 - Spectral analysis (periodograms)
 - Wavelet analysis

$$Y_t = \frac{1}{m} \sum_{j=-k}^k (X_{t+j})$$

Results:

- Fourteen of the fifteen GPS stations show consistent subsidence averaging approximately 2.45 mm/year, while only one station (TWRG) exhibits uplift, likely due to data discontinuity or antenna change.

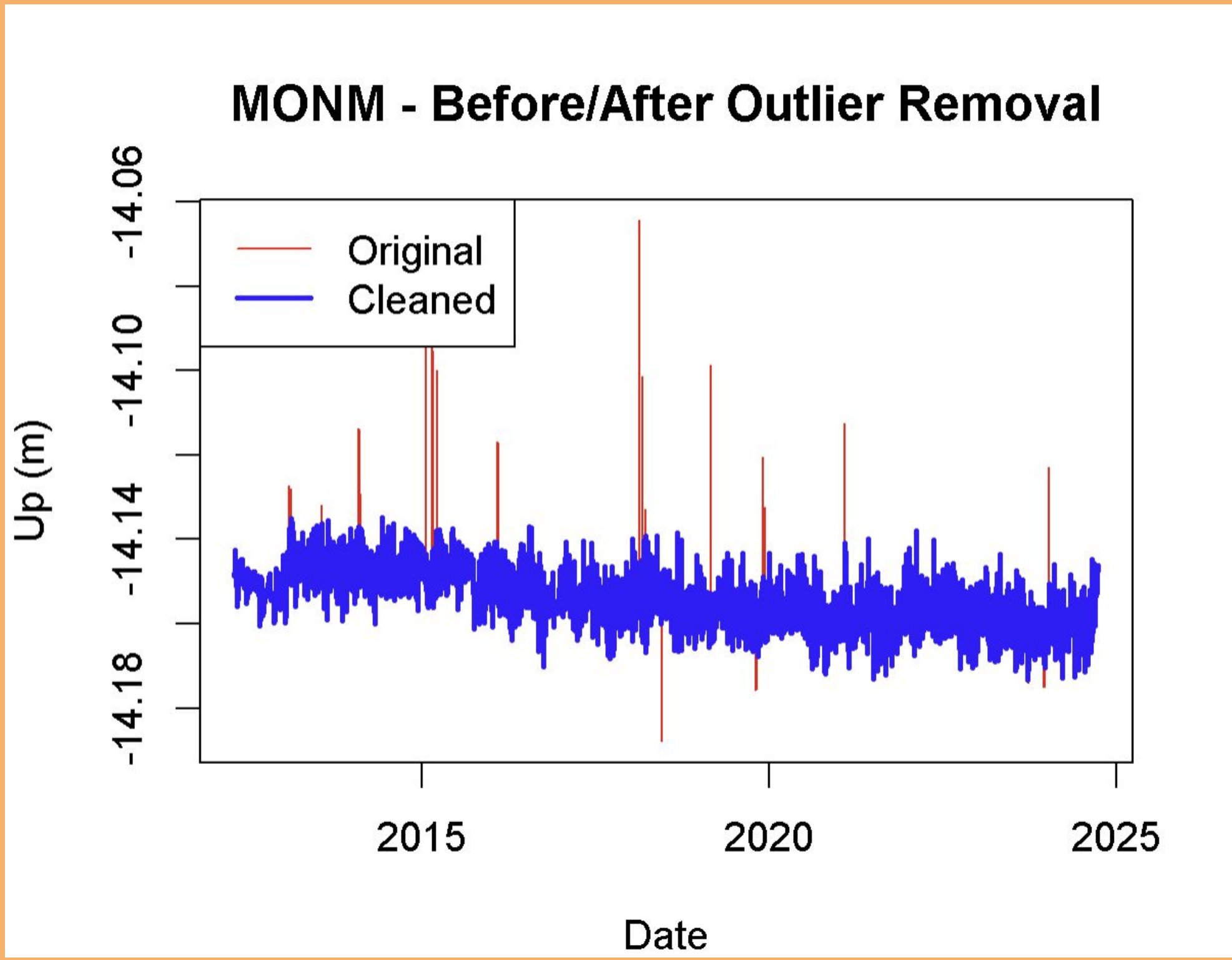


Figure 2: Before and after of outlier removal ($|Z| > 3$) for MONM station

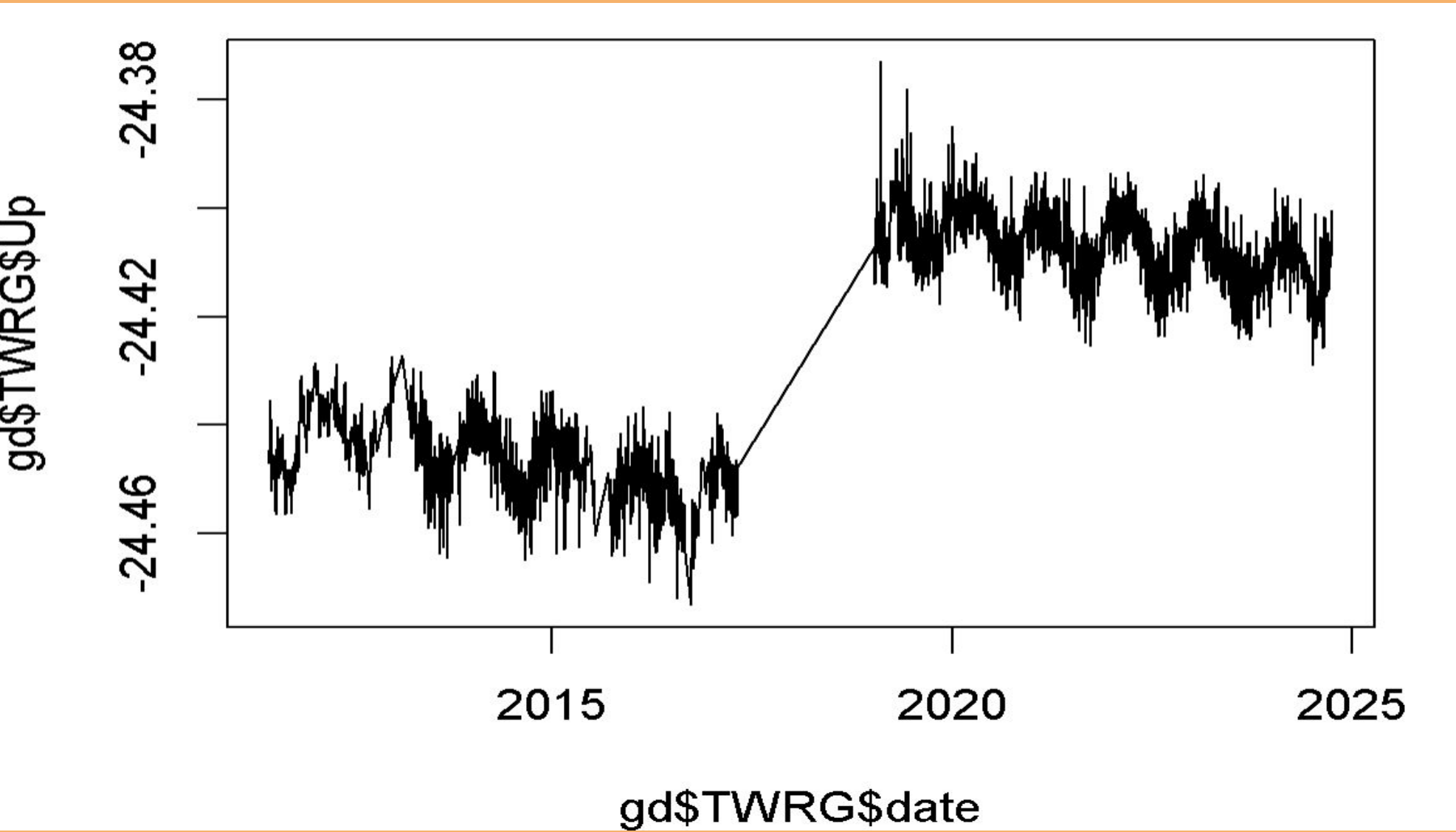


Figure 4: TWRG GPS Up Component (antenna change? or uplift?)

Station	Subsidence Rate	R ²
MONM	-1.17 mm/yr	(R ² =0.888)
NJNT	-1.93 mm/yr	(R ² =0.789)
PLRI	-1.60 mm/yr	(R ² =0.912)
TWMA	-1.99 mm/yr	(R ² =0.938)
NJOC	-0.92 mm/yr	(R ² =0.887)
NJBL	-0.86 mm/yr	(R ² =0.205)
BHVN	-1.07 mm/yr	(R ² =0.969)
NJGT	-0.86 mm/yr	(R ² =0.801)
TWEH	-1.27 mm/yr	(R ² =0.940)
NJAC	-1.43 mm/yr	(R ² =0.985)
NJA1	-2.06 mm/yr	(R ² =0.974)
NJSW	-1.41 mm/yr	(R ² =0.997)
NJCM	-1.29 mm/yr	(R ² =0.949)
CMAF	-16.38 mm/yr	(R ² =0.311)
Average Subsidence	-2.45 mm/yr	
TWRG	4.00 mm/yr	R ² =0.680
Average Uplift	4.00 mm/yr	

Table 1: GPS vertical displacement rates calculated from Kolmogorov-Zurbenko filtered (m=365 days, k=3 iterations) time series, showing regional subsidence averaging -1.28 mm/yr (excluding CMAF and TWRG outliers).

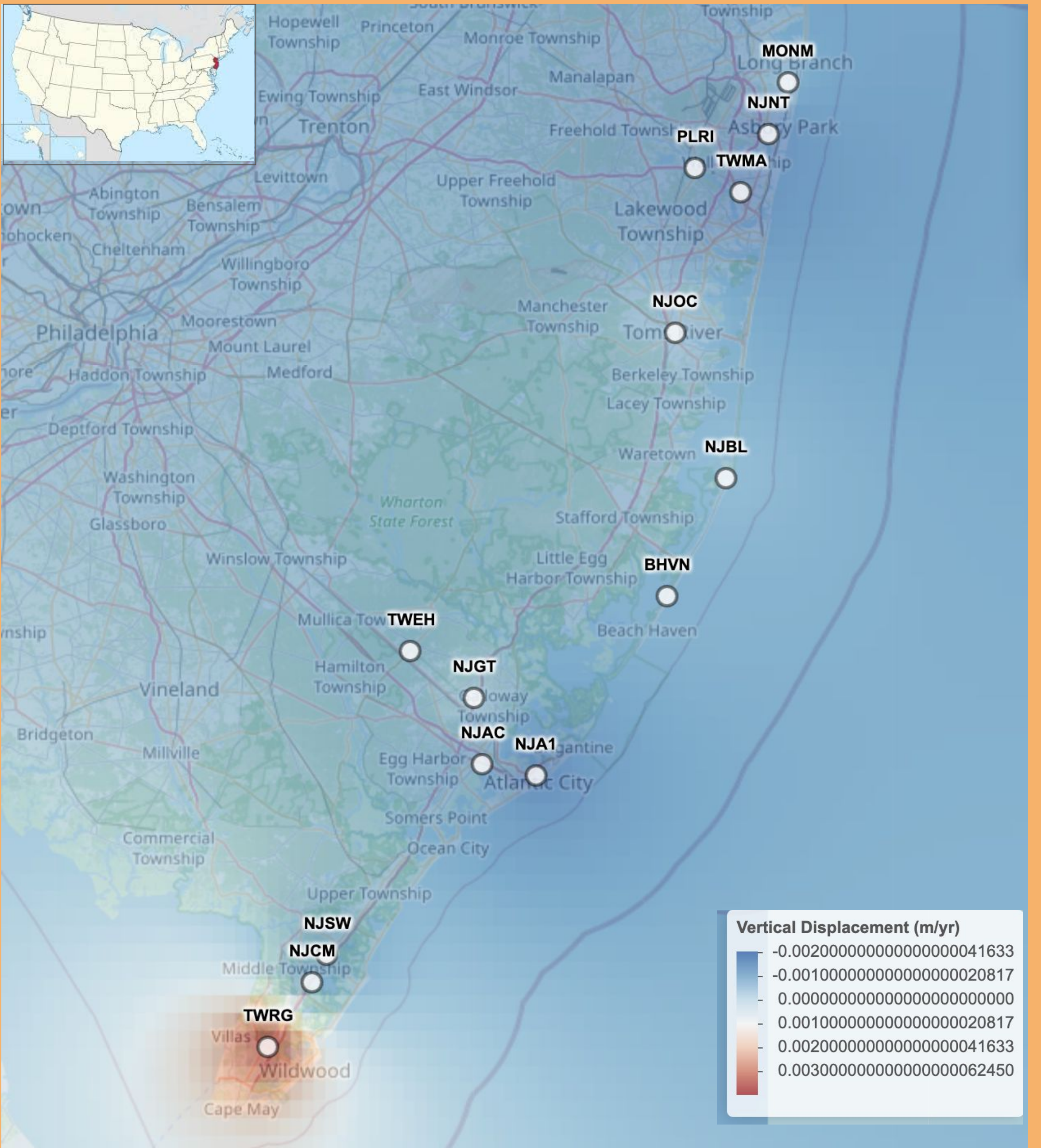


Figure 3: Interpolated GPS vertical displacement rates across coastal New Jersey, with blue indicating subsidence and red indicating uplift (CMAF excluded)

Discussion/Conclusion:

- TWRG is the only station showing uplift. There is a breakpoint of missing years of data (perhaps antenna replacement or station repairs time) (Figure 4)
- The consistent subsidence observed along the New Jersey shoreline suggests ongoing vertical ground lowering that may enhance vulnerability to coastal flooding. Future work will focus on detrending seasonal atmospheric effects and confirming local variability through longer records.
- Observed average subsidence of ≈ 2.45 mm/yr is within the expected range for passive margins affected by glacial isostatic adjustment and sediment compaction. In contrast, tectonically active margins (e.g., Japan or Alaska) commonly exhibit rates of 5–20 mm/yr
- Even moderate subsidence rates of this magnitude can meaningfully amplify relative sea-level rise impacts, emphasizing the need for long-term monitoring and resilience planning.

Acknowledgements:

I would like to thank Dr. Antonios E. Marsellos and the Geology, Environment and Sustainability department at Hofstra University for providing us with the necessary tools to conduct this research.

Credit Authorship Contribution Statement:

Sirichenko D: editing, literature, Rstudio coding, Figures, tables, writing, results, discussion, references, introduction, mapping

Marsellos, A.E.: supervision, Rstudio coding, guidance, mentoring

References:

- Auguie B (2017). gridExtra: Miscellaneous Functions for "Grid" Graphics. R package version 2.3, <https://CRAN.R-project.org/package=gridExtra>
- Barrett T, Dowle M, Srinivasan A, Gorecki J, Chirico M, Hocking T (2024). data.table: Extension of data frame. R package version 1.15.4, <https://CRAN.R-project.org/package=data.table>
- Chan K, Ripley B (2022). TSA: Time Series Analysis. R package version 1.3.1, <https://CRAN.R-project.org/package=TSA>
- Cheng J, Schloerke B, Karambelkar B, Xie Y (2024). leaflet: Create Interactive Web Maps with the JavaScript 'Leaflet' Library. R package version 2.2.2, <https://CRAN.R-project.org/package=leaflet>
- Close B, Zurbenko I, Sun M (2020). kza: Kolmogorov-Zurbenko Adaptive Filters. R package version 4.1.0.1, <https://CRAN.R-project.org/package=kza>
- Grolemund G, Wickham H (2011). Dates and Times Made Easy with {lubridate}. R package version 1.9.3, <https://CRAN.R-project.org/package=lubridate>
- Hester J, Bryan J (2024). glue: Interpreted String Literals. R package version 1.7.0, <https://CRAN.R-project.org/package=glue>
- Hijmans R (2025). raster: Geographic Data Analysis and Modeling. R package version 3.6.32, <https://CRAN.R-project.org/package=raster>
- Moritz S, Bartz-Bielstein T (2017). {imputeTS: Time Series Missing Value Imputation in R}. R package version 3.3, <https://CRAN.R-project.org/package=imputeTS>
- Ooms J (2024). curl: A Modern and Flexible Web Client for R. R package version 5.2.1, <https://CRAN.R-project.org/package=curl>
- Pebesma E, Bivand R (2005). Classes and methods for spatial data in {R}. R package version 2.2.0, <https://CRAN.R-project.org/package=sf>
- Pebesma EJ (2004). Multivariable geostatistics in {S}: the gstat package. R package version 2.1.4, <https://CRAN.R-project.org/package=gstat>
- Roesch A, Schmidbauer H (2018). WaveletComp: Computational Wavelet Analysis. R package version 1.1, <https://CRAN.R-project.org/package=WaveletComp>
- Wickham H (2016). ggplot2: Elegant Graphics for Data Analysis. R package version 3.5.2, <https://CRAN.R-project.org/package=ggplot2>
- Wickham H, François R, Henry L, Müller K, Vaughan D (2023). dplyr: A Grammar of Data Manipulation. R package version 1.1.4, <https://CRAN.R-project.org/package=dplyr>
- Wickham H, Henry L (2025). purrr: Functional Programming Tools. R package version 1.0.4, <https://CRAN.R-project.org/package=purrr>
- Zeileis A, Grothendieck G (2005). zoo: S3 Infrastructure for Regular and Irregular Time Series. R package version 1.8.12, <https://CRAN.R-project.org/package=zoo>